Semantic Field Detection: Tackling Lexical Polysemy

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Course : DS357 :- Natural Language Processing (NLP)

GitHub:- https://github.com/Jagadeesh9110/NLP-semantic-detection

Automatically identifying the knowledge domain (Medical, Finance, etc.) of a sentence by analyzing its "Semantic Field.

The Problem: Lexical Polysemy – words like "Apple," "Bank," or "Mercury" possess number of distinct meanings in the English language.

Dataset: Open English WordNet 2023 Edition

🔗 **GitHub:** https://github.com/globalwordnet/english-wordnet
🔗 **Official site:** https://wordnet.princeton.edu

Literature Survey

Survey Paper 1 — The Linguistic Foundation

Paper: Recent Trends in Word Sense Disambiguation: A Survey (IJCAI 2021) - Bevilacqua, Pasini, Raganato & Navigli

PDF: https://www.ijcai.org/proceedings/2021/0593.pdf

Page: https://www.ijcai.org/proceedings/2021/593

- **Core Research:** Surveys 40 years of Word Sense Disambiguation (WSD) evolution.
- **The Baseline Problem:** Proves that **Knowledge-based systems** (using static dictionary lookups like my **Approach 1**) hit a performance ceiling due to being context-blind.
- **The Contextual Shift:** Highlights that modern Transformer-based models (like BERT) dominate because they use contextual embeddings.
- **Research Gap:** Identifies that most high-performing models act as **"Black Boxes"**—they provide accuracy but lack transparency in *why* a specific domain was chosen.
- **My Implementation:** Motivates **Approach 2** (Context-Aware) and addresses the gap by adding **Innovation 2** (Explainable AI Tracer)

Literature Survey

Survey Paper 2 – Mathematical Field Theory

Paper-2 :- FIRE: Semantic Field of Words Represented as Non-Linear Functions (NeurIPS 2022) - Du & Tanaka-Ishii, University of Tokyo

PDF:https://proceedings.neurips.cc/paper_files/paper/2022/file/f08223bc8d177df6807811c32f5acfed-Paper-Conference.pdf
OpenReview: https://openreview.net/forum?id=3AxaYRmJ2KY

- **Theory (FIRE):** Argues against treating word meanings as a single point in space.
- **Semantic Fields:** Proposes that words project a **"Field"** that exerts a mathematical "pull" on surrounding terms.
- **Context Resolution:** A word's specific meaning is defined by the aggregate field created by its neighbors.
- **Research Gap:** The paper is purely theoretical and mathematical; it lacks a functional, lightweight implementation that end-users can interact with in real-time.
- **My Implementation:** Directly inspired the **Sliding Window Algorithm** used in **Approach 2**, turning high-level theory into a functional Python module.

Literature Survey

Survey Paper 3 – Safety & Uncertainty

Paper-3 :- Ambiguity Meets Uncertainty: Investigating Uncertainty Estimation for Word Sense Disambiguation (ACL 2023) - Zhu Liu & Ying Liu

PDF: https://aclanthology.org/2023.findings-acl.245.pdf
ACL Anthology: https://aclanthology.org/2023.findings-acl.245/
arXiv: https://arxiv.org/abs/2305.13119

- **System Reliability:** Investigates why standard NLP models often make confident mistakes.
- **The Overconfidence Trap:** Discovered that WSD systems are often **"dangerously overconfident,"** assigning high probability to wrong labels rather than admitting confusion.
- **Uncertainty Quantification:** Recommends that systems should explicitly flag ambiguous inputs.
- **Research Gap:** Identifies the need for uncertainty estimation but does not provide a **Feature-Importance Trace** to show which words specifically caused the confusion.
- **My Implementation:** Inspired **Innovation 2** (Uncertainty-Aware Scoring) and my **60% Threshold** to flag AMBIGUOUS results.

Implementation: Technical Stack & NLP Pipeline

Tech Stack

Python 3.12, NLTK (Text Processing), wn (Open WordNet API)

STEP 01

Tokenization

Regex-based cleaning to handle punctuation and case normalization. Ensures uniform input for the disambiguation engine.

STEP 02

Lemmatization

Uses **WordNetLemmatizer** to convert words to root forms (e.g., "running" → "run").

Crucial for accurate WordNet matching and mapping.

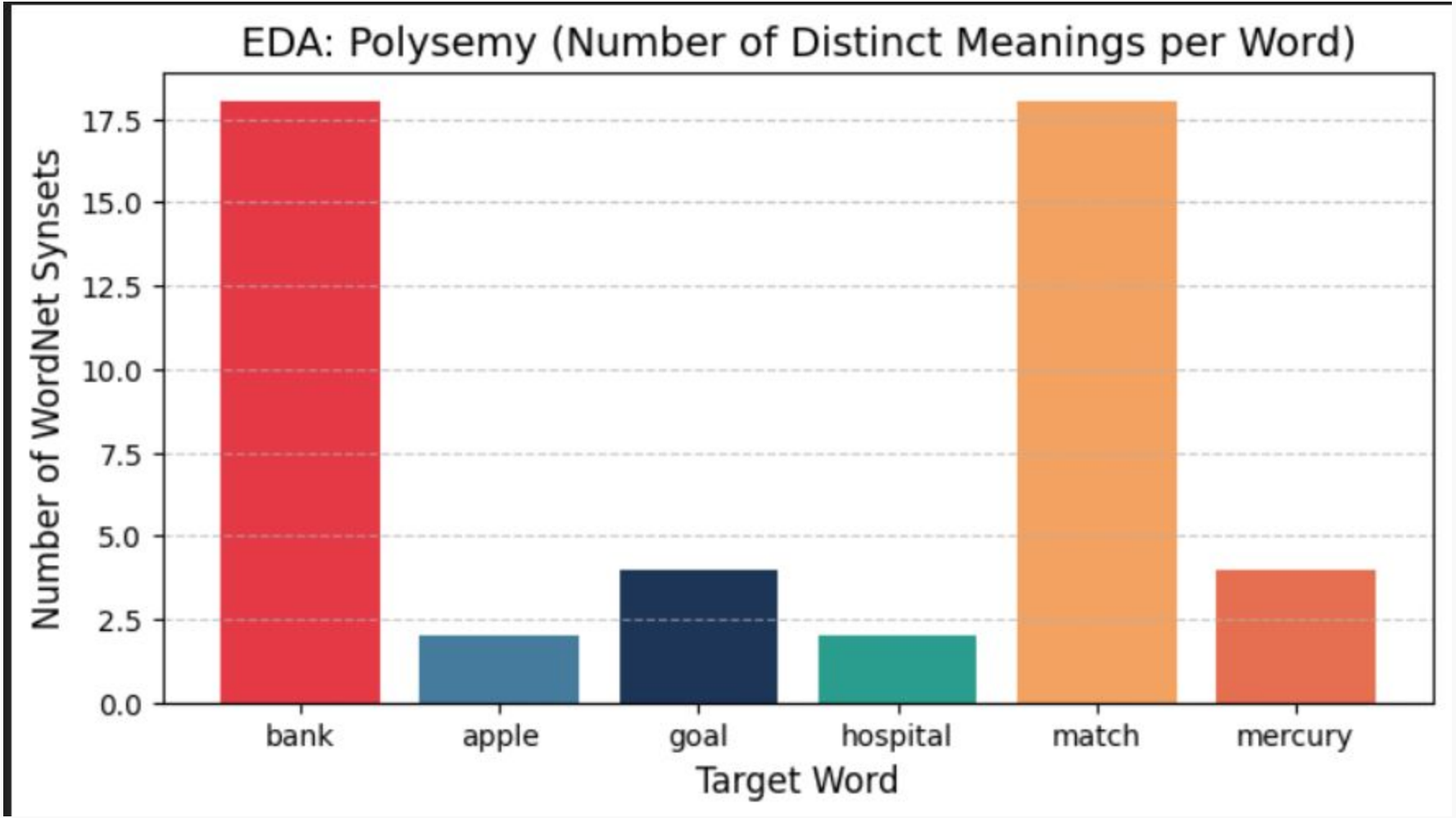
STEP 03

Stop-word Removal

Filtering out high-frequency noise (the, is, at). This allows the model to focus solely on semantic signals.

Implementation: Knowledge Base - Open English WordNet 2023

Metric	Total Count
Total Words	161,338
Total Synsets	120,135
Nouns	123,612
Verbs	11,615
Adjectives	21,619
Adverbs	4,481


EDA: Polysemy (Number of Distinct Meanings per Word)
Number of WordNet Synsets
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
bank
apple
goal
hospital
match
mercury
Target Word

Approach 1: Baseline Dictionary Lookup

CORE LOGIC & MECHANISM

Logic: Sequential token-scan matching keywords against a static dictionary.

Mechanism: Assigns classification to the first available domain signal immediately.

CRITICAL LIMITATIONS

Context-Blindness: Words treated as isolated units; no neighboring information considered.

Polysemy Failure: Incorrectly defaults to first match (e.g., 'Apple' as FOOD).

JUPYTER NOTEBOOK: CONTEXT-AWARE TEST OUTPUT

```
APPROACH 1: Baseline Test

Input : 'Apple released a new phone...'
Tokens : ['apple', 'released', 'new', 'phone', ...]

Result : FOOD
Trigger: 'apple'
```

❌ FAILURE ANALYSIS

The system incorrectly classified a technology sentence as **FOOD** because *'apple'* appeared first in the token sequence.

Critical domain signals like **'phone'** and **'battery'** were entirely ignored due to the sequential "first-match" mechanism.

Approach 2: Sliding Window Algorithm

CORE LOGIC & MECHANISM

Theoretical Root: Implements Distributional Semantics.

Mechanism: For every target word, the model analyzes a context window of +/- 2 words.

DOMAIN VOTING CONSENSUS

Consensus Score: Every keyword in the window casts a "vote" for its mapped domain.

Polysemy Resolution: Aggregate scores determine the winning domain.

JUPYTER NOTEBOOK: CONTEXT-AWARE TEST OUTPUT

```
APPROACH 2: Context-Aware Test

Input : 'Apple released a new phone...'
Tokens : ['apple', 'released', 'new', 'phone',
'better', 'battery']

Result : TECHNOLOGY
Score Board : {'food': 3, 'technology': 11}
Key Drivers : ['apple', 'phone', 'battery']
```

✅ SUCCESS ANALYSIS

The window captured **'phone'** and **'battery'**, which successfully outvoted the food meaning of **'apple'**.

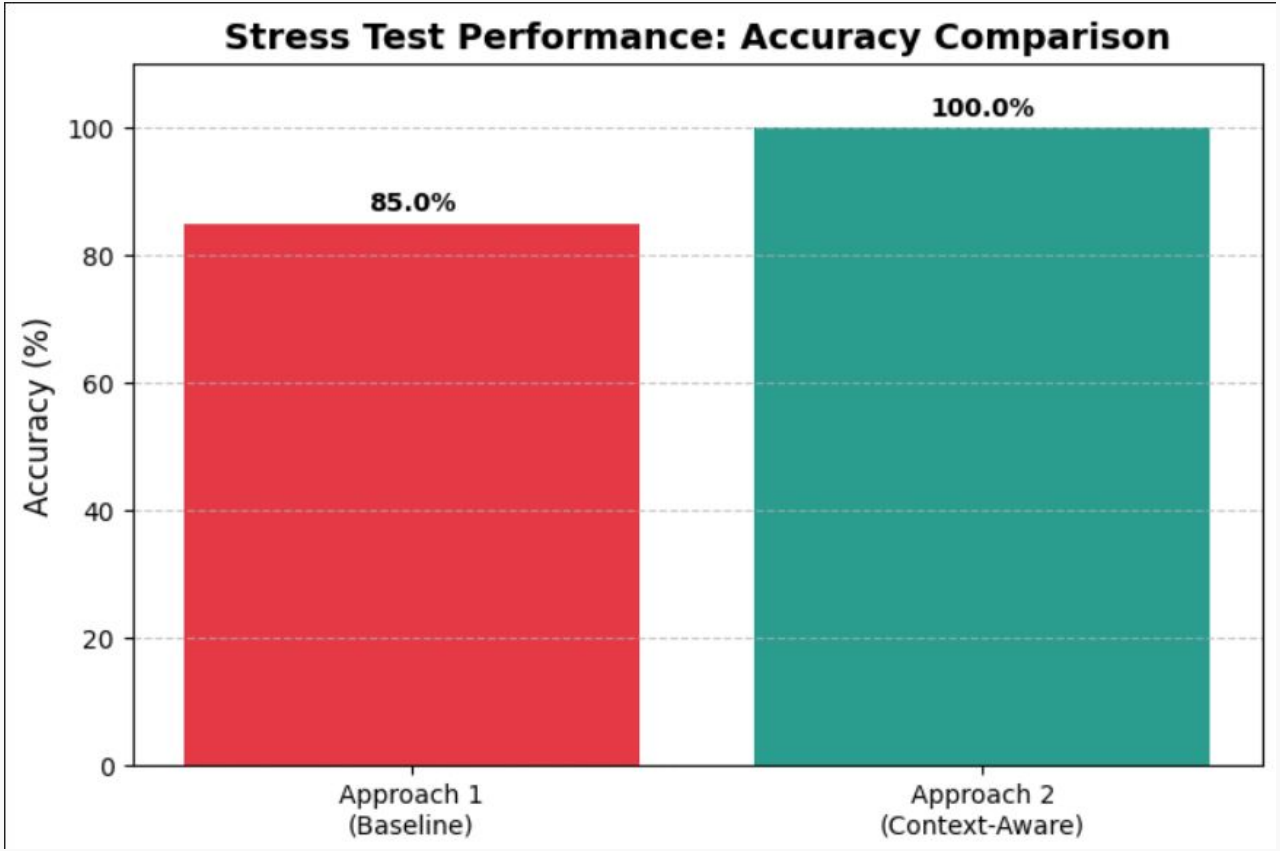
By moving across the tokenized sentence, the algorithm achieves distributional consensus, resolving polysemy through neighborhood context.

Analysis Phase : Head-to-Head Performance Comparison

Sentence	Expected	A1	A2
Apple released a new phone with better bat...	technology	❌	✅
I ate a fresh apple with my meal	food	✅	✅
The doctor performed surgery at the hospit...	medical	✅	✅
Messi scored a brilliant goal in the final	sports	✅	✅
The bank approved a loan at 5 percent inte...	finance	✅	✅
The bank charged high interest on the loan	finance	✅	✅
Ronaldo scored a brilliant goal in the fin...	sports	✅	✅
The doctor gave medicine to the patient in...	medical	✅	✅
I ate an apple with my meal today	food	✅	✅
Apple released software for the new device	technology	❌	✅

```
Approach 1 Accuracy : 8/10 = 80.0%
Approach 2 Accuracy : 10/10 = 100.0%
Improvement         : +20.0% by using context window
```

Analysis: Performance Metrics - Stress Test Accuracy Comparison



Analysis & Interpretative Results

CONCLUSION 1

The baseline system consistently fails when a **polysemous word** (like 'apple') appears before clear domain markers.

CONCLUSION 2

The **+15.0% accuracy gain** is purely attributed to the sliding window's ability to aggregate mathematical consensus.

CONCLUSION 3

Approach 2 proved to be **100% robust** against the specific polysemy traps tested, validating the theoretical "Semantic Field" pull.

FINAL VERDICT

Moving from dictionary lookups to context-aware logic is mandatory for reliable Semantic Field Detection.

Innovation 1: Programmatic Vocabulary Growth via WordNet

THE PROBLEM

Manual domain dictionaries are **static and limited**. They cannot scale to the 161,000+ words available in WordNet, creating a bottleneck for domain-specific NLP tasks.

THE SOLUTION

Leveraging **WordNet synsets and lemmas** to automatically discover related vocabulary from a small set of "seed words," ensuring dynamic scalability.

MECHANISM

The system performs a **graph traversal** from seeds to various senses, extracting all synonyms within synsets to expand the knowledge base automatically.

MEASURED IMPACT: AUTOMATED EXPANSION RESULTS

Finance Domain

3 → 35 Words

profit, interest, currency, etc.

Medical Domain

3 → 26 Words

diagnosis, treatment, symptom, etc.

```
INNOVATION 1 — WordNet Auto-Expansion of Domain Keywords
================================================================

MEDICAL domain:
  Seed words  : ['doctor', 'hospital', 'medicine']
  After WordNet expansion: 26 related words found
  Sample: ['hospital', 'medicament', 'doctor up', 'dr.', 'infirmary', 'doctor of the church', 'furbish up', 'doc']

TECHNOLOGY domain:
  Seed words  : ['computer', 'software', 'phone']
  After WordNet expansion: 27 related words found
  Sample: ['estimator', 'software package', 'data processor', 'telephone set', 'reckoner', 'speech sound', 'telephone', 'package']

FINANCE domain:
  Seed words  : ['bank', 'money', 'invest']
  After WordNet expansion: 35 related words found
  Sample: ['indue', 'banking concern', 'depository financial institution', 'look', 'empower', 'endue', 'reckon', 'count']

SPORTS domain:
  Seed words  : ['football', 'goal', 'athlete']
  After WordNet expansion: 8 related words found
  Sample: ['goal', 'end', 'destination', 'athlete', 'football', 'football game', 'jock', 'finish']

FOOD domain:
  Seed words  : ['apple', 'meal', 'fruit']
  After WordNet expansion: 7 related words found
  Sample: ['repast', 'fruit', 'orchard apple tree', 'yield', 'apple', 'meal', 'malus pumila']
```

Innovation 2: Uncertainty-Aware Scoring & Decision Tracing

XAI TRACER

Moving away from **"Black Box"** models by providing a **mathematical trace** for every classification. It ensures transparency in automated domain detection.

CONFIDENCE SCORING

Quantifies the consensus of the sliding window:

```
(Domain / Total) × 100
```

Provides a **reliability metric** for every decision.

DECISION TRACE

Explicitly identifies which **keywords** drove the result. E.g., 'phone' contributed **33.3%** to a TECHNOLOGY classification result.

EVALUATION RESULT: "Apple released a new phone with better batteries..."

STATUS

High (78.6%)

Confidence Score

TRACE DATA

Driven by: **'apple'** (33.3%), **'phone'** (33.3%), **'battery'** (33.3%).

Innovation 2: Mitigating Model Overconfidence

LINGUISTIC HONESTY

Inspired by **ACL 2023** research, the system is designed to admit when it is "confused." It prioritizes **accuracy over guessing** in high-stakes NLP.

THE 60% THRESHOLD

If the top domain captures **less than 60%** of the total vote, the system flags the result as:

`AMBIGUOUS`

IMPACT

Increases system **reliability** for production environments where "forced" answers are **dangerous** or misleading to end-users.

EXAMPLE OF MANAGED AMBIGUITY: "The hospital monitor was infected by a computer virus..."

DETECTION

Multi-Domain(52.4%<60%)

MEDICAL & TECHNOLOGY detected

SYSTEM RESPONSE

Ambiguous

Warns user rather than forcing wrong label

Driven by: **'virus'** (33.3%), **'computer'** (33.3%), **'monitor'** (33.3%).

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   SEMANTIC FIELD DETECTION SYSTEM
------------------------------------------------------------
   Author  : Manyam Jagadeeswar Reddy (23BDS033)
   Course  : DS357 - Natural Language Processing (NLP)
   Dataset : Open English WordNet 2023
------------------------------------------------------------
   Domains : MEDICAL | TECHNOLOGY | FINANCE | SPORTS | FOOD
------------------------------------------------------------
   Type any English sentence and press Enter.
   Type 'quit' or 'exit' to stop.
------------------------------------------------------------

  >> The company decided that every new apple device must have a faster processor and more memory.

   Domain     : [ TECHNOLOGY ]
   Confidence : 66.7%
   Status     : High confidence
   XAI Trace  : Decision driven by 'device' (50.0%), 'apple' (50.0%)
------------------------------------------------------------
  >> Scientists found a digital virus in the hospital computer that was monitoring the patient's heart rate.

   Domain     : [ MEDICAL ]
   Confidence : 60.0%
   Status     : High confidence
   XAI Trace  : Decision driven by 'patient' (33.3%), 'hospital' (33.3%), 'virus' (33.3%)
------------------------------------------------------------
  >> If you invest your money in the market today, you might see a high profit despite the current economy.

   Domain     : [ FINANCE ]
   Confidence : 100.0%
   Status     : High confidence
   XAI Trace  : Decision driven by 'market' (20.0%), 'money' (20.0%), 'invest' (20.0%), 'profit' (20.0%), 'economy' (20.0%)
------------------------------------------------------------
  >> The athlete ran toward the goal to score the winning point for the team in the final match.

   Domain     : [ SPORTS ]
   Confidence : 100.0%
   Status     : High confidence
   XAI Trace  : Decision driven by 'athlete' (20.0%), 'match' (20.0%), 'goal' (20.0%), 'team' (20.0%), 'score' (20.0%)
```

Summary of Learnings and Next Steps

PROJECT CONCLUSION

- Solved the **Polysemy** problem using a Context-Aware Sliding Window.
- Boosted accuracy on "trap" sentences from **80% to 100%**.
- Integrated **XAI Tracer** for transparent logic and ambiguity management.

KEY TECHNICAL TAKEAWAY

Local context is the **critical driver** for correct Word Sense Disambiguation.

The professional logic-separation architecture ensures the system is modular, maintainable, and highly interactive.

FUTURE SCOPE: EVOLVING THE ARCHITECTURE

Modern Architectures

Transition to **BERT** or Transformers to capture global sentence dependencies beyond sliding windows.

Global Scaling

Ingest all **45 WordNet files** to expand the lexicon across every linguistic domain.

Thank you